

Online Appendix

Mapping O*NET Tasks to ISCO Occupations using Text Similarity

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A Parameter selection

Parameter values are selected through a multi-phase unsupervised search over weights combination that represent the share of each part in the final embedding for task side and the occupation side. The selection procedure is described here before the validation results (Appendices B–D), as the config selection is performed entirely on unsupervised criteria; validation results are examined afterwards as a consistency check only.

Search procedure

An adaptive iterative sweep is conducted over the five blend weights: w_{soc} , w_{dwa} , w_{isco} , $w_{\text{isco-task}}$, $w_{\text{occ}} \in [0, 1]$. All variants share pre-computed embeddings for the SOC occupation titles, ISCO-08 task descriptions, and ESCO occupation text and skill inventory; the variable cost is only the fast blend and FAISS retrieval step. One feasibility constraint is enforced: $w_{\text{soc}} + w_{\text{dwa}} \leq 0.90$, keeping at least 10% weight on the raw task text in the query vector.

Round 1 (initial grid). Five uniformly-spaced values are placed across each parameter’s full range, yielding a $5^5 = 3,125$ candidate grid reduced to 1,250 feasible configurations after the query-weight constraint.

Rounds 2–6 (adaptive zoom). After each round all five parameters are re-gridded simultaneously using a trust-region strategy based on the current local step h and the current best value b for each parameter:

- *Local grid boundary, not global bound:* expand by one full step, targeting $[b - 2h, b - h, b, b + h, b + 2h]$ (clipped to $[0, 1]$).
- *True global bound:* one-sided refinement with a halved step $h' = \max(h/2, h_{\min})$, where the step-balance floor $h_{\min} = \max_p(h_p)/4$ prevents any one dimension from becoming much finer than the others.
- *Interior:* symmetric refinement with halved step h' , placing five points at $[b - 2h', b - h', b, b + h', b + 2h']$.

The sweep will not converge if any parameter is at a grid boundary or if any parameter has not yet been refined below its initial step size, ensuring at least one interior-zoom pass for each dimension. The sweep terminates

when the score improvement falls below 0.0005 and all parameters are at interior grid points with fine-resolution grids.

The six rounds yield 16,000 candidate specifications in total. The per-round configuration counts decrease as the grid concentrates around the optimum: 1,250 (Round 1), 3,000 (Round 2), 2,375 (Round 3), 3,125 (Round 4), 3,125 (Round 5), and 3,125 (Round 6).

Sweep score

Each candidate is scored by

$$\text{sweep_score} = \frac{3 \cdot \text{cov} + 2 \cdot \text{sim} - 2 \cdot \text{overload} - 2 \cdot \text{gini}}{9},$$

where *cov* is the ISCO-08 coverage share (share of 4-digit ISCO-08 unit groups receiving at least one task assignment), *sim* is the mean cosine similarity of retained task–ISCO links, *overload* is the share of tasks assigned to overloaded ISCO groups (those receiving more links than allowed by the overload cap), and *gini* is the Gini coefficient of the task-count distribution across ISCO groups (Table A1). The formula is applied identically in every round without re-normalisation, so scores are directly comparable across rounds.

Table A1: Sweep score components.

Metric	Weight	Direction
ISCO-08 coverage share	3	maximize
Mean cosine similarity retained	2	maximize
Share of tasks in overloaded ISCO group	2	minimize
Gini coefficient of task count per ISCO group	2	minimize

Selected configuration

The configuration with the highest sweep score across all 16,000 candidates is selected as the production specification: $w_{\text{soc}} = 0.375$, $w_{\text{dwa}} = 0.266$, $w_{\text{isco}} = 0.375$, $w_{\text{isco-task}} = 0.734$, $w_{\text{occ}} = 0.156$ (sweep score 0.3907, labelled **r6_s375_d266_i38_t73_o16**; coverage 99.8%, mean similarity 0.748, overloaded share 2.9%, Gini 0.457). The top configurations are closely clustered, confirming that results are robust to small perturbations in these weights.

The chosen configuration can be read directly from its weights. On the *query side*, the blend is constructed in two sequential steps. First, the raw task embedding and the DWA label embedding are combined: $\text{core} = (1 - w_{\text{dwa}}) \cdot \text{task} + w_{\text{dwa}} \cdot \text{DWA}$. Second, the SOC title embedding is blended in: $\text{query} = (1 - w_{\text{soc}}) \cdot \text{core} + w_{\text{soc}} \cdot \text{title}$. In the implementation, component vectors are L2-normalised before blending and the

query vector is L2-normalised after each blend, so the weights should be interpreted as nominal blend weights rather than exact post-normalisation vector shares. Because the dials are applied sequentially, the effective contributions of each source are: SOC title 37.5% ($= w_{\text{soc}}$), raw task text 45.9% ($= (1 - w_{\text{soc}})(1 - w_{\text{dwa}})$), and DWA labels 16.6% ($= (1 - w_{\text{soc}}) \cdot w_{\text{dwa}}$). All three sources play a meaningful role, with the raw task text providing the largest share of task-specific content, the SOC title anchoring occupational context, and DWA labels supplying a complementary functional signal.

On the *target side*, $w_{\text{isco}} = 0.375$ allocates 37.5% of the target representation to ISCO-08 text and 62.5% to the ESCO component. Within the ISCO component, $w_{\text{isco-task}} = 0.734$ gives 73.4% weight to ISCO task-item descriptions versus 26.6% to the generic group description. Within the ESCO component, $w_{\text{occ}} = 0.156$ allocates only 15.6% to occupation descriptions and 84.4% to the ESCO skill inventory. In nominal target-side weights, the ESCO skill inventory is the single largest contributor to the target embedding (52.7%), followed by ISCO task items (27.5%), ISCO information text (10.0%), and ESCO occupation text (9.8%). The target vector is L2-normalised before retrieval. This configuration indicates that skill-level vocabulary provides the most precise semantic anchor for matching O*NET tasks to ISCO-08 unit groups.

Figures A1 and A2 show the sweep score surface for the target-side and query-side parameters respectively. Each cell reports the maximum sweep score across all configurations sharing that pair of parameter values (marginalising over the remaining parameters). The selected configuration is marked with a red star.

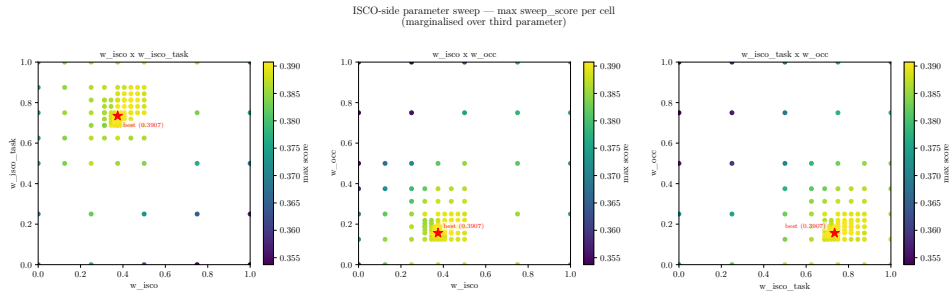


Figure A1: Target-side parameter sweep: maximum sweep score per cell for pairwise combinations of w_{isco} , $w_{\text{isco-task}}$, and w_{occ} (third parameter marginalised). Red star marks the selected configuration.

Query-side (task) parameter sweep — max sweep_score per cell

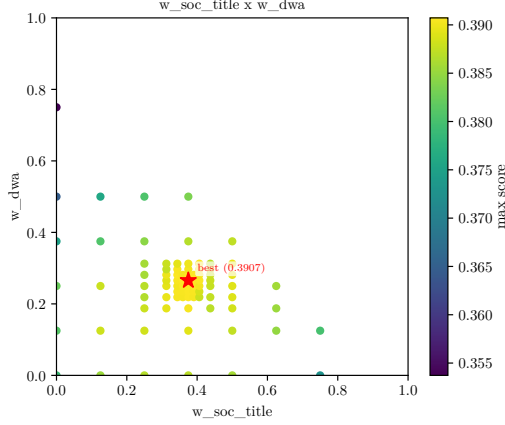


Figure A2: Query-side parameter sweep: maximum sweep score per cell for w_{soc} and w_{dwa} . Red star marks the selected configuration.

Findings

The sweep reveals that DWA labels provide a meaningful positive contribution at $w_{\text{dwa}} \approx 0.27$, which translates to an effective contribution of $\approx 16.6\%$ of the query vector: neither zero nor dominant DWA weight is optimal, indicating that functional task descriptors complement but do not replace the raw task text and SOC title anchors.

On the target side, the ESCO skill inventory dominates the optimal representation (effective weight $\approx 52.8\%$ of the full target vector), while ISCO task descriptions contribute $\approx 27.5\%$. This indicates that skill-level vocabulary bridges the gap between O*NET task language and ISCO-08 unit groups more precisely than the official ISCO-08 task descriptions alone. The relatively low weight on ISCO-08 text ($w_{\text{isco}} = 0.375$) reflects the sparseness of the ISCO-08 task lists for some unit groups, which limits their discriminative power when used alone.

Each task is assigned to a single ISCO-08 unit group. Allowing multiple assignments increases coverage but reduces interpretability and complicates aggregation of task-based measures. High-performing configurations cluster in a narrow region of the parameter space, indicating that results are robust to reasonable variation in parameter values.

B Validation framework

Validation compares occupation-level assignments implied by the task-based mapping to institutional SOC–ISCO concordances constructed by statistical agencies and international organisations. Task-level assignments are first ag-

gregated to the SOC occupation level. Each SOC occupation is assigned the ISCO-08 unit group receiving the largest number of task matches. Agreement is then evaluated relative to institutional SOC–ISCO mappings at both the four-digit unit-group level and the one-digit major-group level.

Multiple institutional concordances are used in order to avoid dependence on any single mapping source. Agreement is therefore evaluated relative to both strict and lenient benchmarks. The strict benchmark requires agreement with all available crosswalks for a given SOC occupation, while the lenient benchmark requires agreement with at least one institutional source. Results are similar across benchmark definitions, indicating that agreement rates are not driven by any single concordance.

Institutional crosswalks are used only for validation and do not influence construction of the task-level mapping.

C Full validation results

Table C1 reports chain crosswalk validation results for alternative benchmark definitions and O*NET releases. The main text reports results for O*NET 29.2 using the lenient benchmark, which balances measurement error across institutional concordances.

Table C1: Chain crosswalk validation: match rates across scenarios.

Scenario	Cov. (%)	Exact (%)	Sub- major (%)	Major group (%)
<i>SOC 2018 (O*NET 29.2)</i>				
A1 – ESCO-SOC18	99.9	67.7	80.9	88.3
A2 – SOC18-ESCO	98.7	59.7	74.0	81.5
A3 – Strict intersection	98.5	59.6	73.8	81.5
A4 – Lenient union	99.9	68.0	81.2	88.4
<i>SOC 2010 (O*NET 25.0)</i>				
B1 – ESCO-ONET-MHV ^a	79.1	50.2	65.9	74.8
B2 – SOC10-ISCO-BLS	99.9	44.3	59.8	68.6
B3 – Lenient union (MHV $\cup$ BLS)	100.0	49.4	66.1	74.2
B4 – Strict intersection (MHV $\cap$ BLS)	70.1	49.3	63.1	71.2
^a Less independent: derived via the same semantic similarity approach as the pipeline.				

Table C2 reports occupation-level agreement for each dataset and institutional crosswalk separately. Pair precision and recall measure the overlap between the set of ISCO unit groups implied by the pipeline (one per SOC occupation) and the set of unit groups listed in the institutional crosswalk.

SOC in-set rate is the share of SOC occupations whose pipeline-implied top-1 ISCO falls anywhere in the set of acceptable ISCO codes listed for that SOC in the institutional crosswalk, reported at three levels of the ISCO hierarchy: exact 4-digit unit group, 2-digit sub-major group, and 1-digit major group. For many-to-many concordances with flat scores (the ESCO–O*NET–SOC concordance), which map each SOC to multiple equally-weighted ISCOs with no preferred choice, in-set rate is the appropriate measure: comparing against a single arbitrarily tie-broken reference code would understate agreement by construction.

Table C2: Occupation-level agreement with institutional crosswalks.

Dataset	Institutional crosswalk	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)	Shared SOC n
O*NET 29.2	ESCO-ONET-SOC (SOC18)	0.343	0.464	0.395	0.868	0.911	0.944	797
O*NET 29.2	ONET-SOC-ESCO (SOC18)	0.247	0.608	0.352	0.812	0.862	0.898	784
O*NET 25.0	ESCO-ONET (MHV)	0.131	0.703	0.221	0.733	0.811	0.852	576
O*NET 25.0	BLS ISCO-SOC	0.151	0.715	0.249	0.678	0.756	0.804	774

Table C3 reports internal consistency between pairs of institutional crosswalks for the same SOC version. High within-version agreement between independent reference sources indicates that the institutional benchmarks themselves are coherent.

Table C3: Internal consistency between pairs of institutional crosswalks.

SOC version	Reference A	Reference B	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)
SOC 2010	ESCO-ONET (MHV)	BLS ISCO-SOC	0.613	0.543	0.576	0.810	0.856	0.895
SOC 2018	ESCO-ONET-SOC (SOC18)	ONET-SOC-ESCO (SOC18)	0.544	0.988	0.701	0.598	0.690	0.738

Figure C1 illustrates SOC in-set rate across the institutional crosswalks for both O*NET releases.

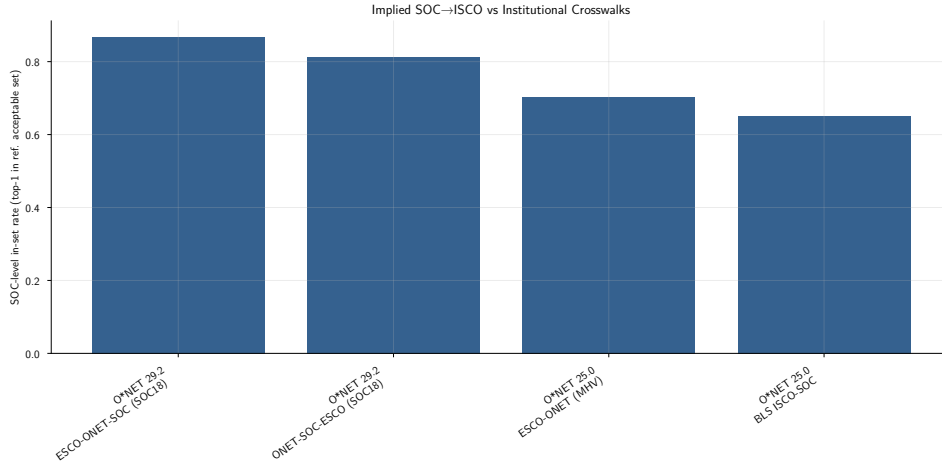


Figure C1: SOC in-set rate with institutional crosswalks by O*NET release (share of SOC occupations whose pipeline top-1 ISCO falls within the crosswalk’s acceptable set).

Table C4 lists occupations where the pipeline-implied ISCO code disagrees with the institutional crosswalk. These results should be treated as suggestive rather than definitive, because disagreement can originate from the reference crosswalk rather than from the pipeline. A concrete illustration comes from the ESCO–O*NET semantic crosswalk (Matysiak et al. 2024, SOC10): for *Advertising Sales Agents* (SOC 41-3011) the pipeline assigns ISCO 2431 (*Advertising and Marketing Professionals*), which is a direct task-content match. That concordance, however, maps this occupation to 3339 (*Business Services Agents NEC*) and 9510 (*Street and Related Service Workers*) — a pairing that appears to reflect noise in the underlying semantic similarity rather than expert judgement. In cases such as this the pipeline assignment is arguably more accurate than the benchmark, so the validation in-set rate provides a lower bound on true classification quality.

Table C4: Examples of occupations where the pipeline disagrees with an institutional crosswalk.

Dataset	Reference	SOC	SOC title	Implied ISCO	Reference ISCO	Task support	Support share
O*NET 29.2	ONET-SOC-ESCO (SOC18)	19-3022	Survey Researchers	4227 Survey and Market Research Interviewers	2120 Mathematicians, Actuaries and Statisticians	16	1.000
O*NET 29.2	ONET-SOC-ESCO (SOC18)	11-3131	Training and Development Managers	2421 Training and Staff Development Professionals	1212 Human Resource Managers	12	1.000
O*NET 29.2	ONET-SOC-ESCO (SOC18)	43-9081	Proofreaders and Copy Markers	4413 Coding, Proofreading and Related Clerks	2642 Journalists	11	1.000
O*NET 29.2	ONET-SOC-ESCO (SOC18)	17-3031	Surveying and Mapping Technicians	2165 Cartographers and Surveyors	3112 Civil Engineering Technicians	29	0.987
O*NET 29.2	ONET-SOC-ESCO (SOC18)	19-3051	Urban and Regional Planners	2164 Town and Traffic Planners	2422 Policy Administration Professionals	24	0.960
O*NET 29.2	ONET-SOC-ESCO (SOC18)	13-1151	Training and Development Specialists	2421 Training and Staff Development Professionals	1212 Human Resource Managers	19	0.950
O*NET 29.2	ONET-SOC-ESCO (SOC18)	27-2041	Music Directors and Composers	2622 Musicians, Singers and Composers	2641 Authors and Related Writers	28	0.933
O*NET 29.2	ONET-SOC-ESCO (SOC18)	53-7071	Gas Compressor and Gas Pumping Station Operators	3134 Petroleum and Natural Gas Refining Plant Operators	3133 Chemical Processing Plant Controllers	12	0.923
O*NET 29.2	ONET-SOC-ESCO (SOC18)	33-9091	Crossing Guards and Flaggers	5419 Protective Services Workers Not Elsewhere Classified	8312 Railway Brake, Signal and Switch Operators	11	0.917
O*NET 29.2	ONET-SOC-ESCO (SOC18)	49-9092	Commercial Divers	7541 Underwater Divers	3123 Construction Supervisors	21	0.913
O*NET 29.2	ONET-SOC-ESCO (SOC18)	13-1111	Management Analysts	2423 Management and Organization Analysts	2413 Financial Analysts	10	0.909
O*NET 29.2	ONET-SOC-ESCO (SOC18)	29-3031	Radiologic Technologists and Technicians	3211 Medical Imaging and Therapeutic Equipment Technicians	2209 Health Professionals Not Elsewhere Classified	27	0.900
O*NET 25.0	BLS ISCO-SOC	19-3022	Survey Researchers	4227 Survey and Market Research Interviewers	2120	16	1.000
O*NET 25.0	BLS ISCO-SOC	11-3131	Training and Development Managers	2421 Training and Staff Development Professionals	1212	12	1.000
O*NET 25.0	BLS ISCO-SOC	41-3041	Travel Agents	4221 Travel Consultants and Clerks	3339	8	1.000
O*NET 25.0	BLS ISCO-SOC	13-1151	Training and Development Specialists	2421 Training and Staff Development Professionals	2396	19	0.950
O*NET 25.0	BLS ISCO-SOC	47-3053	Helper- Plumbers, Pipefitters, and Steamfitters	7126 Plumbers and Pipe Fitters	9313	14	0.933
O*NET 25.0	BLS ISCO-SOC	19-3099	Transportation Planners	2164 Town and Traffic Planners	2633	20	0.909
O*NET 25.0	BLS ISCO-SOC	53-3022	Airfield Operations Specialists	3154 Air Traffic Controllers	3151	19	0.905
O*NET 25.0	BLS ISCO-SOC	25-3099	Tutors	2399 Teaching Professionals Not Elsewhere Classified	2353	17	0.905
O*NET 25.0	BLS ISCO-SOC	13-2051	Financial Analysts	2413 Financial Analysts	2412	16	0.899
O*NET 25.0	BLS ISCO-SOC	43-4131	Loan Interviewers and Clerks	3112 Credit and Loans Officers	4312	16	0.889
O*NET 25.0	BLS ISCO-SOC	47-5051	Rock Splitters, Quarry	7113 Stonemasons, Stone Cutters, Splitters and Carvers	8111	8	0.889
O*NET 25.0	BLS ISCO-SOC	53-7032	Excavating and Loading Machine and Dragline Operators	8302 Earthmoving and Related Plant Operators	8111	14	0.875

D Author annotation of O*NET 29.2 task assignments

To complement the automated chain-crosswalk validation, 108 O*NET 29.2 tasks were selected and annotated by the author. Tasks were drawn from 36 SOC occupations covering all nine ISCO major groups, with occupations chosen in order of combined US and Nordic employment within each group and tasks selected by O*NET importance score (top three per occupation). For each task the author assigned the most appropriate ISCO-08 four-digit unit group from a shortlist of plausible candidates, without reference to the pipeline’s predictions.

The evaluation below serves as a post-hoc consistency check on the unsupervised config selection performed in Appendix A. The selected configuration ($w_{\text{soc}} = 0.375$, $w_{\text{dwa}} = 0.266$, $w_{\text{isco}} = 0.375$, $w_{\text{isco-task}} = 0.734$, $w_{\text{occ}} = 0.156$, max links= 1) achieves 36.1% exact four-digit agreement with the author annotations, rising to 57.4% at the two-digit sub-major level and 72.2% at the one-digit major-group level, at a mean cosine similarity of 0.739. On the automated chain-crosswalk benchmark (lenient union A4), the selected configuration achieves 68.0% exact agreement.

These results confirm that the semantic similarity approach produces assignments broadly consistent with expert judgement at the occupation-group level. The author annotation reflects a single annotator’s judgements and should be interpreted as an approximate consistency check rather than a definitive gold standard; the automated chain-crosswalk benchmark, which is based on institutional SOC–ISCO concordances, is the primary validation criterion.

E Coverage and example matches

The mapping assigns O*NET tasks to 435 of the 436 ISCO-08 unit groups. The only group without assigned tasks is ISCO 7516 (Tobacco Preparers and Tobacco Products Makers), which has no functional equivalent in O*NET task descriptions.

Coverage is broad across major occupational groups, including professional, technical, clerical, service, and manual occupations. Differences in task counts across ISCO-08 groups reflect differences in the level of detail in O*NET task descriptions and structural differences between U.S. and international occupational classifications.

Figure E1 shows how coverage, mean similarity, and overload share evolve across the five pipeline stages for both O*NET releases. The retrieval stage (S1) achieves broad coverage at the cost of many candidates per task; subsequent filtering stages progressively tighten precision while maintaining near-complete coverage.

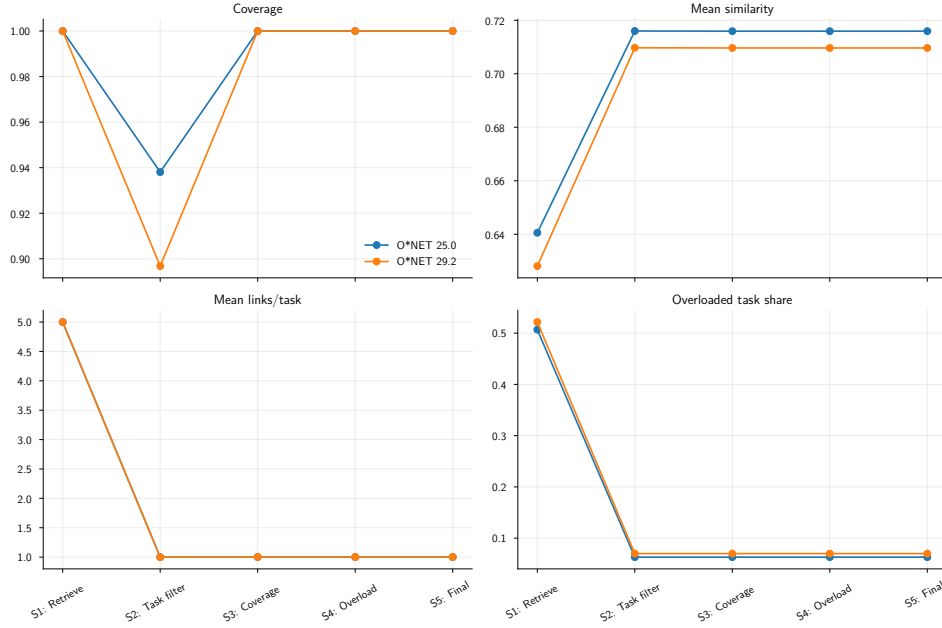


Figure E1: Pipeline metrics at each processing stage (S1–S5) for both O*NET releases.

Table E1 reports the corresponding numeric values for each stage and dataset.

Table E1: Pipeline metrics by stage and dataset.

Dataset	Stage	Coverage	Mean similarity	Mean links/task	Overloaded task share
O*NET 25.0	S1: Retrieve	0.998	0.689	5.000	0.467
O*NET 25.0	S2: Task filter	0.968	0.745	1.000	0.049
O*NET 25.0	S3: Coverage	0.998	0.745	1.001	0.049
O*NET 25.0	S4: Overload	0.998	0.745	1.001	0.049
O*NET 25.0	S5: Final	0.998	0.745	1.001	0.049
O*NET 29.2	S1: Retrieve	1.000	0.628	5.000	0.522
O*NET 29.2	S2: Task filter	0.897	0.710	1.000	0.070
O*NET 29.2	S3: Coverage	1.000	0.710	1.002	0.070
O*NET 29.2	S4: Overload	1.000	0.710	1.002	0.070
O*NET 29.2	S5: Final	1.000	0.710	1.002	0.070

Figure E2 compares final-stage (S5) summary metrics across both O*NET releases.

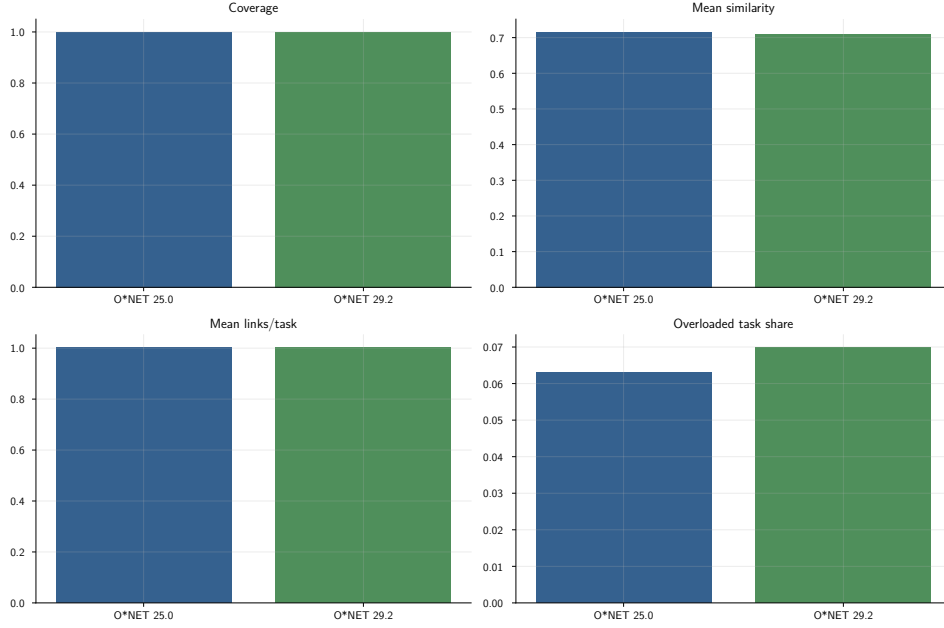


Figure E2: Final-stage (S5) metrics for O*NET 29.2 and O*NET 25.0.

Table E2 summarises the S5 metrics numerically.

Table E2: Final-stage (S5) summary metrics by dataset.

Dataset	Coverage	Mean similarity	Mean links/task	Overloaded task share	Gini
O*NET 25.0	0.998	0.745	1.001	0.049	0.447
O*NET 29.2	1.000	0.710	1.002	0.070	0.488

Table E3 lists the ISCO-08 unit groups that attract the largest numbers of task assignments before the overload-control step (S4). These groups tend to be broad “not elsewhere classified” categories whose descriptions overlap with many O*NET tasks.

Table E3: ISCO-08 groups with the highest task counts at S3 (before overload control).

Dataset	ISCO	Occupation label	Tasks at S3	Tasks at S4	Tasks at S5	Pruned in S4
O*NET 29.2	2310	University and Higher Education Teachers	363	363	363	0
O*NET 29.2	7223	Metal Working Machine Tool Setters and Operators	269	269	269	0
O*NET 29.2	2149	Engineering Professionals Not Elsewhere Classified	242	242	242	0
O*NET 29.2	3211	Medical Imaging and Therapeutic Equipment Technicians	223	223	223	0
O*NET 29.2	2131	Biologists, Botanists, Zoologists and Related Professionals	216	216	216	0
O*NET 29.2	2352	Special Needs Teachers	199	199	199	0
O*NET 29.2	3131	Power Production Plant Operators	196	196	196	0
O*NET 29.2	2133	Environmental Protection Professionals	183	183	183	0
O*NET 25.0	2310	University and Higher Education Teachers	289	289	289	0
O*NET 25.0	1345	Education Managers	271	271	271	0
O*NET 25.0	7233	Agricultural and Industrial Machinery Mechanics and Repairers	207	207	207	0
O*NET 25.0	7223	Metal Working Machine Tool Setters and Operators	206	206	206	0
O*NET 25.0	2351	Education Methods Specialists	195	195	195	0
O*NET 25.0	5329	Personal Care Workers in Health Services Not Elsewhere Classified	188	188	188	0
O*NET 25.0	3211	Medical Imaging and Therapeutic Equipment Technicians	186	186	186	0
O*NET 25.0	2359	Teaching Professionals Not Elsewhere Classified	186	186	186	0

Table E4 reports illustrative task assignments, one per ISCO major group, showing the full query-side inputs (task text, SOC occupation, DWA labels) alongside the pipeline assignment.

Table E4: Examples of task assignments (one per ISCO major group). O*NET task text, SOC occupation title, and the top two DWA activity labels constitute the query-side inputs; the ISCO-08 assignment is the pipeline output.

O*NET Task	SOC Occupation	DWA Labels (top 2)	ISCO-08 Assignment
Develop and implement organizational policies and procedures for the facility or medical unit	Medical and Health Services Managers	Develop operating strategies, plans, or procedures.; Develop organizational policies or programs	1342 Health Service Managers
Prepare, analyze, or verify annual reports, financial statements, and other records, using accepted accounting and statistical procedures to assess financial condition and facilitate financial planning	Accountants and Auditors	Prepare financial documents, reports, or budgets.; Verify accuracy of financial information	2411 Accountants
Obtain and verify evidence by interviewing and observing suspects and witnesses or by analyzing records	Detectives and Criminal Investigators	Examine records or other types of data to investigate criminal activities.; Interview people to gather information about criminal activities	3355 Police Inspectors and Detectives
Process and issue employee paychecks and statements of earnings and deductions	Payroll and Time-keeping Clerks	Execute sales or other financial transactions	4313 Payroll Clerks
Monitor conduct of prisoners in housing unit, or during work or recreational activities, according to established policies, regulations, and procedures, to prevent escape or violence	Correctional Officers and Jailers	Maintain surveillance of individuals or establishments	5413 Prison Guards
Feed and water animals, and clean and disinfect pens, cages, yards, and hutches	Animal Breeders	Care for animals.; Clean equipment or facilities	6129 Animal Producers Not Elsewhere Classified
Operate newly repaired machinery or equipment to verify the adequacy of repairs	Industrial Machinery Mechanics	Test mechanical equipment to ensure proper functioning	7233 Agricultural and Industrial Machinery Mechanics and Repairers
Observe and interpret readings on gauges, meters, and charts registering various aspects of boiler operation to ensure that boilers are operating properly	Stationary Engineers and Boiler Operators	Monitor equipment operation to ensure proper functioning	8182 Steam Engine and Boiler Operators
Sort materials, such as metals, glass, wood, paper or plastics, into appropriate containers for recycling	Recycling and Reclamation Workers	Sort recyclable materials	9612 Refuse Sorters